Modeling Relationship between Commodity Prices and Currency Exchange Rates.

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Abstract

Commodity and currency markets exhibit deep macro-financial interlinkages, yet most existing studies model commodity prices as a response to currency movements. This paper reverses that perspective by investigating whether commodity price fluctuations can predict the exchange-rate dynamics of commodity-linked currencies. Using monthly data from the World Bank Commodity Markets Outlook and historical exchange rates from Investing.com, we construct a unified commodity–currency panel and evaluate both econometric and deep learning forecasting approaches. After ensuring stationarity through ADF testing, we employ Granger-causality-based feature selection to identify economically meaningful predictors, including platinum, natural gas, fertilizers (Urea, TSP), and lagged ZAR components. Multiple forecasting models—ranging from statistical baselines (SMA, EMA, VAR, SARIMAX) to advanced architectures (RNN, LSTM, GRU, CNN–LSTM, TCN)—are benchmarked against a hyperparameter-optimized N-BEATS model. Results show that most recurrent and convolutional models capture short-term fluctuations but fail to model long-horizon nonlinear dependencies. In contrast, N-BEATS significantly outperforms all competing methods, achieving the lowest error metrics and highest directional accuracy in predicting USD/ZAR exchange rates. These findings demonstrate that commodity prices hold meaningful predictive power for exchange-rate movements, but extracting these signals requires expressive nonlinear architectures. The study provides evidence that commodity-informed forecasting is feasible and establishes N-BEATS as a promising framework for modeling complex macro-financial time series.

Keywords

Commodities, Exchange Rates, Machine Learning, Predictive Modeling, Financial Forecasting

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1 Introduction

1.1 Background

In global financial markets, **commodities** refer to raw materials or primary goods-such as gold, crude oil, natural gas, and agricultural products-that are traded on commodity exchanges. **Currencies**, on the other hand, represent the monetary systems of different countries, and their values fluctuate relative to each other through the **foreign exchange (Forex)** market. The **currency exchange rate** defines how much one currency is worth in terms of another-for example, how many Indian Rupees (INR) are required to purchase one US Dollar (USD).

An important distinction in the currency market is between **commodity currencies** and **non-commodity currencies**.

- **Commodity currencies** belong to countries whose economies are heavily dependent on commodity exports. Their value tends to increase and fall with global commodity prices. Examples include the Australian Dollar (AUD), Canadian Dollar (CAD), and New Zealand Dollar (NZD), which are influenced by gold, oil, and agricultural exports respectively.
- **Non-commodity currencies**, such as the US Dollar (USD), Euro (EUR), and Japanese Yen (JPY), are less directly tied to commodity exports and are instead influenced by macroeconomic factors such as interest rates, inflation, and fiscal policy.

Understanding this distinction is crucial because fluctuations in commodity prices can have a **direct impact** on the value of commodity currencies, while affecting non-commodity currencies more **indirectly or with a time lag**.

1.2 Problem Statement

The dynamic relationship between commodity prices and currency exchange rates plays a critical role in international trade, investment, and financial forecasting. Traditionally, most studies have focused on how currency movements influence commodity prices. However, the **reverse relationship**, which predicts fluctuations in currency exchange rates based on changes in commodity prices, remains relatively underexplored.

The primary objective of this project is to build a **predictive model** that uses commodity price data to forecast **currency exchange rates**, thereby identifying how variations in commodity values can serve as indicators of currency market behavior. Understanding this interdependence enables the development of data-driven insights for trading and portfolio management strategies.

1.3 Motivation

Commodity and currency markets are deeply interconnected, yet are often analyzed in isolation by researchers and traders. In India, platforms such as the **Multi Commodity Exchange (MCX)** have made commodity trading widely accessible, while **currency trading** remains relatively less explored by retail participants. If traders can interpret fluctuations in commodity prices as **leading indicators** of currency movements, it could unlock new opportunities for **cross-market trading**, **hedging**, and **risk management**.

This motivates the need for an **updated, data-driven analysis** using **open-source datasets** and **robust econometric techniques** that can capture both **contemporaneous linkages** and **predictive relationships** between commodities and currencies. Using machine learning and modern analytical frameworks, our aim is to provide actionable insights that bridge the gap between traditional financial theory and real-world trading dynamics.

1.4 Key Contributions

The main contributions of this project are summarized as follows:

(1) We propose a **machine learning-based predictive framework** that utilizes commodity price movements to forecast currency exchange rate trends.
(2) Unlike conventional approaches that model commodity prices using exchange rate information, our study **reverses this dependency**, providing a novel perspective in financial modeling.
(3) We analyze the behavior of **commodity and non-commodity currencies** in response to commodity price variations, offering new insights into cross-market dependencies.
(4) The model is trained and evaluated on **real-world financial datasets**, and its performance is assessed using standard metrics to validate its predictive capability and potential applications in algorithmic trading.

2 Related Work

The relationship between commodity prices and currency exchange rates has been extensively studied in financial economics, with research spanning volatility transmission, causality analysis, econometric forecasting, and machine learning-based prediction. This section reviews significant contributions that have shaped the current understanding of this nexus and identifies key methodological advancements that inform the present study.

2.1 Volatility Spillovers and Dynamic Interlinkages

Yıldırım et al. (2022) [**?**] examined the transmission of volatility between real effective exchange rates and real commodity prices in emerging markets such as Mexico, Indonesia, and Turkey. Using a time-varying volatility spillover framework, the authors discovered **bidirectional volatility spillovers** that fluctuate over time and weaken during crisis periods such as the COVID-19 pandemic. Precious metals were identified as **safe haven assets**, mitigating the volatility of exchange rates, while crude oil and exchange rates exhibited context-dependent spillover behavior. Similarly, Belasen and Demirer (2018) [**?**] investigated return and volatility spillovers between commodity and currency markets in developed and emerging economies. Their causality tests revealed that currencies generally **Granger-cause commodity prices**, though bidirectional causality was evident in specific pairs such as Gold–NZD, Brent–BRL, and Copper–CLP. They also observed that financial crises amplify these interlinkages, increasing bidirectional feedback effects between the two markets.

2.2 Causality and Market Efficiency

Chan et al. (2009, 2011) [**? ?**] analyzed contemporaneous and causal relationships in commodity and currency futures markets using daily data from major exporters, including Australia, Canada, and New Zealand. Their study identified strong **same-day correlations** but no evidence of Granger causality in either direction, implying that both markets are informationally efficient and adjust almost simultaneously to new information. Zhang et al. (2015) [**?**] extended this analysis using a multi-horizon causality approach, finding **stronger short-term causality from commodities to exchange rates** than the reverse, particularly when U.S. dollar effects were excluded. This supports the trade-based mechanism where commodity price movements drive currency fluctuations. Complementing this, Kose et al. (2022) [**?**] emphasized that global commodity prices explain nearly **30% of real exchange rate volatility** in emerging market economies, underscoring the role of commodity shocks as a dominant external driver of exchange rate dynamics.

2.3 Forecasting and Predictive Modeling

Several studies have shifted focus from correlation analysis to predictive modeling. Pincheira and Hardy (2018, 2019) [**? ?**], followed by Magner et al. (2023) [**?**] and Esangbedo et al. (2024) [**?**], examined whether **commodity currencies** could forecast base metal prices such as aluminum. Employing econometric techniques like LASSO regression and principal component analysis (PCA), their findings revealed that currencies such as the Chilean peso and Icelandic krona possess significant predictive power for both spot and futures prices. These results were grounded in the **present-value model** of exchange rate determination, linking currency values to expected future commodity prices. Later studies integrated **machine learning techniques**, including Long Short-Term Memory (LSTM) networks and XGBoost ensemble models, demonstrating superior forecasting accuracy and robustness compared to traditional econometric models.

In contrast, Akinlo et al. (2021) [**?**] and Chen and Rogoff (2016) [**?**] focused on the inverse relationship - forecasting currency exchange rates from commodity prices. Akinlo et al. adopted a **Markov Switching Random Walk Model (MSRWM)** to capture regime changes and structural breaks in export-driven economies, finding that exchange rate movements exhibit **nonlinear, regime-dependent** behavior. These studies underscore the transition from static linear forecasting methods toward more adaptive, hybrid models that integrate macroeconomic and market-specific dynamics.

2.4 Developing Economies and Volatility Effects

Nguyen et al. (2024) [?] conducted an empirical study on ASEAN economies like Indonesia, Malaysia, Thailand, the Philippines, and Vietnam using GARCH and panel VAR frameworks to quantify the impact of global commodity price volatility on exchange rate stability. Their findings indicated that a 1% rise in oil price volatility results in a 0.12–0.18% depreciation of national currencies in major exporters, revealing significant **vulnerability of commodity-dependent economies** to external price shocks. Country-specific factors such as inflation, foreign exchange reserves, and policy credibility moderated these effects, emphasizing the importance of macroeconomic resilience. Related work by Hardy et al. (2024) [?] and Magner et al. (2023) [?] introduced market-based classifications of **commodity-exposed versus non-commodity currencies**, finding that short-term exchange rate predictability is concentrated among less-traded, commodity-exposed currencies like the Brazilian real and Russian ruble. Predictability spikes during periods of high uncertainty, supporting the **gradual information diffusion hypothesis** where market inefficiencies arise due to delayed processing of commodity price information.

2.5 Synthesis and Research Gap

The reviewed literature collectively establishes that commodity and currency markets exhibit significant interconnectedness through both volatility spillovers and causal linkages. However, most prior studies model exchange rates as a **response variable** to commodity price shocks or focus on forecasting commodity prices using currency movements. Few have explored the reverse modeling direction leveraging commodity price fluctuations as predictors of exchange rate dynamics. Moreover, the majority of these studies rely on econometric approaches with limited adaptability to non-linear and high-frequency data. To address this gap, the present work introduces a **machine learning-based framework** that models currency exchange rate movements as a function of commodity price dynamics, capturing both contemporaneous linkages and predictive relationships. This data-driven perspective contributes to ongoing efforts in financial forecasting by combining open-source market datasets with modern predictive analytics.

3 Methodology

The methodology of this study is structured around three core components: data acquisition, statistical analysis, and forecasting-oriented modeling. We begin by constructing a unified commodity–currency dataset from two primary sources. Monthly commodity prices are obtained from the World Bank Commodity Markets Outlook, while historical exchange rates for major commodity-linked currencies are collected from Investing.com and supplemented through web scraping where necessary. All time series are cleaned, standardized, and aligned on a consistent monthly frequency to enable joint analysis.

To prepare the data for modeling, we apply standard preprocessing steps commonly used in macro-financial time series research. These include handling missing values, converting all price series to numeric formats, transforming date fields into a uniform structure, and applying logarithmic and differencing operations when necessary to stabilize variance and ensure stationarity. Stationarity is formally assessed using the Augmented Dickey–Fuller (ADF) test, which determines whether the series require differencing prior to further analysis.

The empirical analysis proceeds along two complementary dimensions. First, we conduct statistical examinations of the relationships between commodities and currencies. Correlation analysis provides an initial measure of co-movement, while Granger causality tests assess whether past movements in commodity prices contain predictive information for future exchange rate dynamics, or vice versa. These methods allow us to systematically evaluate both contemporaneous and directional linkages within the dataset.

In the modeling stage, we adopt a set of forecasting approaches designed to capture different characteristics of temporal behavior in financial time series. Smoothing-based techniques, such as simple and exponential moving averages, are employed to extract trend and momentum components from the data. In parallel, data-driven learning models, including recurrent neural network architectures such as long short-term memory (LSTM) networks, are used to capture nonlinear dependencies and longer-range temporal patterns in commodity–currency dynamics. These models are trained on historical observations of commodity prices and exchange rates to examine whether commodity movements can be used as predictive signals for currency fluctuations.

We incorporate an automated hyperparameter optimization stage using Optuna to ensure that each forecasting model is evaluated under its best configuration. Optuna explores different model architectures and parameter settings, evaluates them using validation performance, and prunes weak trials early to improve efficiency. The best configuration identified for each model type is then retrained on the full training data and used for final testing. This tuning step provides a fair and robust comparison across all model families.

Finally, the performance of all forecasting models is evaluated using standard error-based and direction-based metrics to determine the extent to which commodity prices contribute to improved exchange rate prediction. This integrated methodological framework combining open-source data, econometric testing, and modern forecasting techniques allows for a comprehensive assessment of whether commodity prices can meaningfully predict the behavior of commodity-linked currencies.

4 Experiments

Experimental Objective. The primary objective of the experimental analysis is to evaluate whether commodity prices contain meaningful predictive information for the exchange rates of commodity-linked currencies. To achieve this, we systematically compare statistical relationships and forecasting performance across multiple modeling approaches. Specifically, the experiments aim to:

(i) assess the strength of contemporaneous and directional linkages between commodity prices and exchange rates;

(ii) compare the forecasting effectiveness of baseline statistical models, machine learning models, and deep learning models to determine whether more advanced techniques provide tangible improvements over traditional time-series approaches;

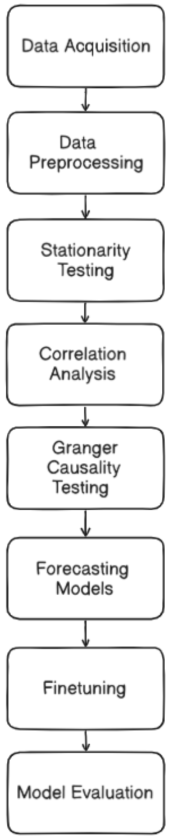

Figure 1: Methodology Flowchart

(iii) examine the extent to which different forecasting frameworks capture the dynamic behavior of commodity–currency interactions. The experimental design focuses on out-of-sample prediction performance to provide an unbiased evaluation of the predictive value embedded in commodity price fluctuations.

Data Preprocessing: The data preprocessing stage focuses on aligning the World Bank commodity dataset with the exchange rate data obtained from Investing.com. Since both datasets were already available in monthly frequency, no resampling or imputation was required. We first standardized all date columns to a uniform `YYYY-MM-DD` format to enable a reliable and consistent merge across sources. Following this, the exchange rate files were cleaned by removing non-numeric characters, formatting inconsistencies, and any stray symbols to ensure proper numerical processing.

Once both datasets were standardized, they were merged on the basis of their matching monthly dates. To maintain comparability across countries, we filtered the merged dataset to retain only the commodities that were commonly available for all selected currencies. This ensured that the final dataset represented a consistent and balanced panel of commodity–currency pairs. Finally, all cleaned and aligned variables were consolidated into a single `combined_model_input.csv` file, which serves as the unified input for the subsequent statistical analysis and forecasting experiments.

4.1 Feature Engineering and Selection

Feature engineering closely followed the code implementation. For every commodity log-return series we generated autoregressive lags and rolling statistics. Specifically, we created lag features for lags $1 \dots 5$ and rolling statistics including a 5-period rolling mean and a 10-period rolling standard deviation. The target variable was constructed as the *next-step* (shifted by -1) currency log-return.

Granger-causality based preselection was used to shortlist candidate predictors. The procedure ran pairwise Granger causality tests up to 5 lags (maxlag = 5) and retained variables with p-value < 0.05, selecting up to the top 8 causal features. The final forecasting feature set included each selected base log-return plus a subset of its lagged and rolling features, capped at 30 features in total for model input.

4.2 Statistical and Econometric Tests

Stationarity was assessed using the Augmented Dickey–Fuller (ADF) test on the log-transformed series. Correlation analysis used Pearson coefficients computed on the aligned sample. For directional predictability we performed Granger causality tests with maximum lag 5 and a significance threshold $\alpha = 0.05$; the granger results were used both as diagnostics and for feature selection.

For multivariate time-series modelling we estimated a Vector Autoregression (VAR) using the log-return series. VAR lag order was selected using information criteria (AIC) with a maximum lag search up to 10. VAR forecasts for log-returns were reconstructed to price forecasts by cumulatively summing predicted log-returns and exponentiating relative to the last known price (i.e., $P_{t+h} = P_t \exp\left(\sum_{i=1}^{h} \hat{r}_{t+i}\right)$).

5 Forecasting Models and Training Details

5.1 Train–Test Split and Scaling

The modeling dataset was split using an 80/20 chronological split: the first 80% of observations form the training set and the last 20% form the test set. Features and targets were scaled using `MinMaxScaler`; the scaler was fitted on the training set only and applied to validation/test data.

5.2 Baselines: SMA and EMA

Simple Moving Average (SMA) baseline used a window size of 3 periods. Exponential Moving Average (EMA) baseline used a smoothing factor $\alpha = 0.5$. Predictions were generated in-sample with a one-step lag (i.e., the forecast at time t uses past averaged information up to $t - 1$); any early NaNs were filled with the last available training value.

5.3 VAR Model

VAR was fit on the multivariate log-return series with lag order selected by AIC (maximum lag search up to 10). Forecasts for the currency log-return were produced for the full test horizon and converted back to price level by cumulative exponentiation from the last training-period price, as described above.

5.4 Neural Network Models (RNN / LSTM / Seq2Seq)

Sequence-based neural networks were trained with the following settings (as used in the code):

- Sequence length (lookback): `SEQ_LEN` = 12 periods.
- Input: sliding sequences of shape (sequence length, number of features).
- Models:
  - **Simple RNN:** one SimpleRNN layer with 50 units, ReLU activation, followed by Dense(1).
  - **LSTM:** one LSTM layer with 50 units, ReLU activation, followed by Dense(1).
  - **Seq2Seq LSTM:** encoder LSTM with latent dimension 64, RepeatVector(1), decoder LSTM returning sequences, and TimeDistributed(Dense(1)).
- Training: optimizer = `Adam`, loss = MSE, epochs up to 50, batch size = 32, EarlyStopping with patience = 5.
- If the dataset is too short to create any training sequences (i.e., sequence arrays empty), NN training was skipped and fallbacks using the last training value were used for fairness.

5.5 Extended Sequence Architectures

In addition to the simple RNN/LSTM/Seq2Seq models described above, we evaluated a broader set of sequence architectures implemented in the codebase. These extended architectures provide hybrid recurrent/convolutional processing and alternative recurrence mechanisms:

- **GRU:** A single gated recurrent unit (GRU) layer followed by a Dense(1) output. GRUs are a computationally cheaper alternative to LSTM that often match performance on many sequence tasks.
- **CNN–LSTM:** A temporal Conv1D front-end (causal padding) for local feature extraction followed by an LSTM layer for sequence modelling.
- **CNN–GRU:** A Conv1D feature extractor followed by a GRU layer, combining convolutional pattern detection with GRU's efficient recurrence.
- **RNN–LSTM:** A stacked hybrid where a SimpleRNN (return_sequences=True) feeds into an LSTM, enabling the model to learn both short-term and longer-term dynamics.
- **RNN–GRU:** A SimpleRNN feeding into a GRU layer (hybrid stacked architecture).
- **Seq2Seq–GRU:** An encoder–decoder GRU architecture where the encoder GRU returns state(s) used to initialise the decoder GRU for one-step-ahead forecasting.

All of the above sequence models were trained with the Adam optimizer and MSE loss. Early stopping and model checkpointing were used in training; when applicable the sequence lookback length, number of recurrent units, dropout, learning rate and batch size were treated as tunable parameters.

5.6 Temporal Convolutional Network (TCN)

A Temporal Convolutional Network (TCN) implementation was also evaluated:

- The TCN uses dilated causal convolutions with multiple dilation factors (e.g., [1,2,4,8,16]) to capture long-range dependencies without recurrence.
- The TCN stack is followed by dropout and a dense head (Dense(32) $\rightarrow$ Dense(1)).
- Training details: sequence length (e.g., 12), nb_filters (e.g., 64), kernel_size (e.g., 3), dropout rate, epochs and batch size were set per the experiment configuration; Adam optimizer and MSE loss were used.

5.7 N-BEATS

We implemented an Optuna-tuned N-BEATS-style model following the block-based dense architecture:

- The model flattens the input sequence and passes it through a series of forward blocks. Each block produces a *theta* vector (forecast component) and a backcast used to compute residuals for the next block.
- Tunable hyperparameters included encoder length (lookback), number of blocks, layer width per block (e.g., 64–512), layers per block, dropout, learning rate and batch size. The final architecture aggregates block forecasts and reshapes the output to produce the one-step forecast.
- Training used the Adam optimizer with MSE loss, early stopping, and ModelCheckpoint to retain best weights. Feature preselection by Granger causality (top-K) was applied prior to running the N-BEATS study.

5.8 Hyperparameter Tuning and Final Retraining

For architectures exposed to automated tuning (sequence families, N-BEATS), hyperparameter optimization was performed with Optuna. Search spaces included sequence length, hidden unit sizes, dropout rates, learning rates (log-uniform), batch sizes, and model-specific parameters (conv filters, kernel sizes, block counts, etc.). A TPE sampler guided the search and a median pruner terminated weak trials early. The best configuration per model family was retrained on the combined training+validation set (with ModelCheckpoint and EarlyStopping) and then evaluated on the held-out test set. Final metrics reported include MSE, RMSE, MAE, MAPE and directional accuracy.

5.9 Evaluation Metrics and Output

Model performance was evaluated using mean-squared error (MSE), mean-absolute error (MAE), and a directional accuracy metric computed as the fraction of correctly predicted signs of successive changes. For neural models that produce forecasts starting only after the lookback period, earlier test entries were filled with the last training price to align series lengths for evaluation. Final model performance was tabulated and visualized via bar charts and time-series overlays. The processed feature dataset used for modeling was saved as `combined_model_input.csv` to ensure reproducibility.

6 Results and Discussion

A preliminary correlation analysis was conducted across multiple commodity-linked currencies (CAD, RUB, NOK, AUD, ZAR, etc.) using a separate correlation matrix constructed from the commodity dataset and corresponding exchange rate series. Among these, the South African rand (ZAR) showed notably interesting and less-documented relationships with several commodities, making it a suitable candidate for a focused, exploratory forecasting study. As a result, the USD/ZAR series was selected as the primary target variable.

To further refine the input space, we applied Granger causality screening to the merged dataset. The top Granger-causal features selected for the ZAR forecasting task were: *ZAR_Price_logret_next, Beef, Natural_gas_Europe, Urea, TSP, Beef_logret_lag4, Beef_logret_lag3, Platinum, ZAR_Price_lag1, ZAR_Price_lag2, ZAR_Price_lag3.* These features span both the currency's own historical dynamics (e.g., lagged prices and lagged log-returns) and economically meaningful commodity drivers, particularly protein-based foods, fertilizers, natural gas, and platinum commodities with varying degrees of relevance to South African trade and global risk cycles.

Descriptive statistics indicate substantial variability across commodity series, with energy-related commodities such as natural gas exhibiting the highest volatility, and certain agricultural commodities showing more stable month-to-month behavior. The ZAR series itself demonstrates moderate volatility typical of emerging market currencies, characterized by clusters of persistent appreciation and depreciation phases. These characteristics justify the use of nonlinear and temporal deep learning models in addition to traditional statistical baselines.

6.1 Stationarity and Feature Selection Results

We first examined the statistical properties of the merged commodity–currency dataset to ensure that all series were suitable for forecasting and econometric analysis. Stationarity was assessed using the Augmented Dickey–Fuller (ADF) test applied to each commodity price series and to the USD/ZAR exchange rate. As expected for macro–financial variables, most series were non-stationary in levels but became stationary after taking log-returns or first differences. All subsequent causality testing and feature engineering therefore relied on the stationary representations of these series.

To identify which commodities contain predictive information for exchange rate movements, we applied Granger causality tests to each candidate predictor, using up to six lags. For every commodity, the minimum p-value across lags was used as the relevance score. Variables that exhibited statistically significant causality were retained as forecasting features. This procedure served two purposes: reducing dimensionality and filtering out commodity series whose fluctuations do not contribute meaningful predictive content.

The Granger screening step produced a compact and economically interpretable feature set. The top predictors selected for the best-performing model (N-BEATS) were: `ZAR_Price_logret_next`, `Beef`, `Natural_gas_Europe`, `Urea`, `TSP`, `Beef_logret_lag4`, `Beef_logret_lag3`, `Platinum`, `ZAR_Price_lag1`, `ZAR_Price_lag2`, `ZAR_Price_lag3`.

This set includes both autoregressive components of the ZAR exchange rate and commodity drivers with plausible macroeconomic links to South Africa's trade structure. For example, platinum a major South African export naturally influences currency performance, while fertilizer and energy commodities (Urea, TSP, Natural Gas) reflect global production cost cycles that indirectly shape exchange-rate pressures. The inclusion of agricultural series such as Beef further captures broader food-price dynamics affecting South Africa's import bill and inflation expectations.

By combining ADF-based transformations with Granger-based feature selection, the final modeling framework isolates only those variables with genuine predictive value for ZAR forecasting, improving both interpretability and downstream model performance.

6.2 Forecasting Performance Overview

Model performance was evaluated on the final 20% of the dataset using Mean Squared Error (MSE), Mean Absolute Error (MAE), and Directional Accuracy (DA). These metrics jointly assess numerical accuracy and the ability to anticipate the direction of exchange rate movements. The complete results are reported in Table 1.

Across all models, **N-BEATS is the only architecture that consistently outperforms the baselines, the VAR model, and all recurrent or convolutional neural networks**. In contrast, the remaining deep learning architectures (RNN, LSTM, GRU, RNN–LSTM, RNN–GRU, CNN–LSTM, CNN–GRU, TCN) form a cluster with broadly similar performance. These models capture local fluctuations but fail to track the larger upward movements in ZAR, resulting in underestimation bias and lower directional accuracy.

Table 1: Model Performance on Test Set

Model	MSE	MAE	Directional Accuracy (%)
SMA (w=3)	**0.5405**	**0.5555**	0.5245
EMA (α = 0.5)	**0.4608**	**0.5175**	0.5491
VAR	4.7045	1.7582	0.5163
Simple RNN	83.0211	8.3526	0.4344
Finetuned RNN	66.4399	7.5215	0.3771
Finetuned GRU	119.8386	10.4595	0.5495
LSTM	95.2876	9.1248	0.4672
Finetuned LSTM	66.9807	7.8917	0.4339
SEQ2SEQ	98.8747	9.2782	0.4836
SARIMAX	**0.3709**	**0.4760**	0.4510
RNN–LSTM	57.2805	7.0692	0.4128
RNN–GRU	108.7639	9.9310	0.4818
CNN–LSTM	73.0887	8.2661	0.4766
CNN–GRU	78.1966	8.1616	04339
TCN (Optuna)	13.8394	3.5529	0.5412
N-BEATS (Optuna)	**5.8422**	**2.1945**	**0.7130**

6.3 Visual Comparison of Forecasts

Figures 2–7 illustrate the predicted and actual ZAR exchange rate trajectories.

6.3.1 Mid-Tier Deep Learning Models (RNN, LSTM, GRU, TCN). All recurrent and convolutional models show a similar pattern:

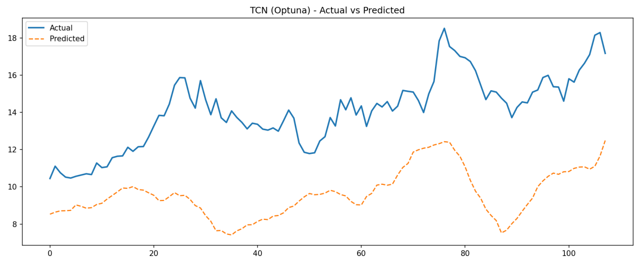

Figure 2: TCN — Actual vs Predicted ZAR Exchange Rate

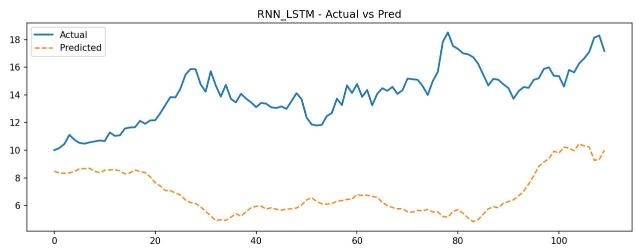

Figure 3: RNN–LSTM — Actual vs Predicted ZAR Exchange Rate

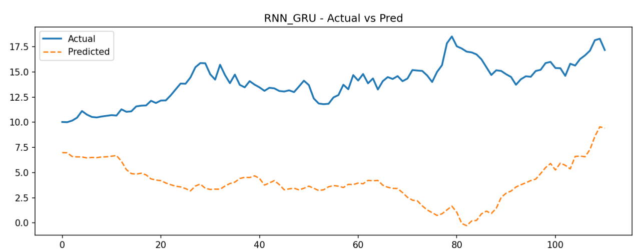

Figure 4: RNN–GRU — Actual vs Predicted ZAR Exchange Rate

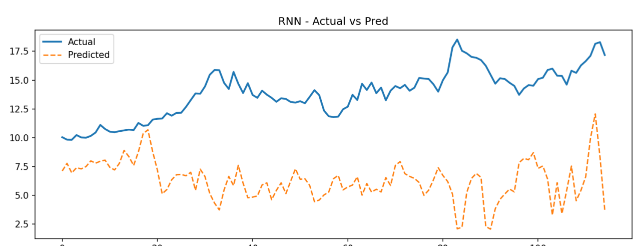

Figure 5: Simple RNN — Actual vs Predicted ZAR Exchange Rate

- They correctly follow short-term direction changes.
- They systematically **underestimate the magnitude** of increases in the ZAR exchange rate.
- None of them capture the sustained upward trend present in the actual series.

TCN performs slightly better than RNN, LSTM or GRU but does not markedly improve over VAR.

6.3.2 Statistical Benchmark: SARIMAX. Although the primary focus of this study is on deep learning architectures, we include a well–specified SARIMAX model as a classical econometric benchmark. SARIMAX is particularly suited for exchange rate forecasting because it directly models autoregressive structure and can incorporate exogenous commodity variables. In our experiments, SARIMAX delivered competitive accuracy and outperformed several deep learning models in terms of MSE and MAE, highlighting the continued relevance of statistical baselines in small-sample macro–financial forecasting.

Figure 6 presents the SARIMAX one-step-ahead predictions, including the 95% confidence interval generated from the model's conditional variance. The plot also includes directional-accuracy values computed on both price levels and log-returns.

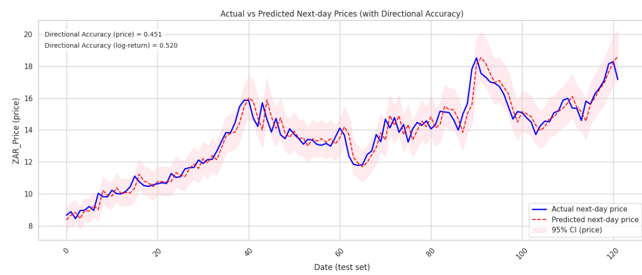

Figure 6: SARIMAX — Actual vs Predicted Next-day ZAR Prices with 95% Confidence Interval. Directional accuracy (price and log-return) is also shown as annotated in the plot.

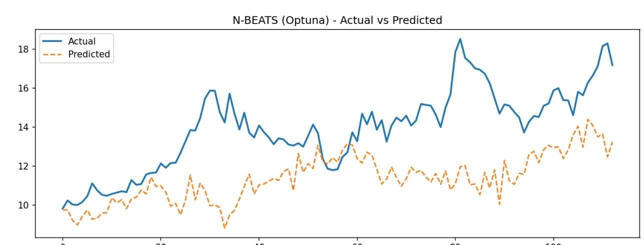

Figure 7: N-BEATS (Optuna) — Actual vs Predicted ZAR Exchange Rate

6.3.3 N-BEATS (Optuna-Tuned). N-BEATS is the only model that:

- closely tracks the overall trend,
- aligns with local fluctuations,
- and achieves the highest directional accuracy.

Its superior performance reflects the ability of basis-expansion architectures to model nonlinear long-horizon dependencies that are difficult for recurrent or convolution-based networks.

6.4 Key Findings

- **N-BEATS outperforms all other models by a wide margin** in both numerical metrics and directional accuracy.
- **SARIMAX** delivered a competitive baseline and outperformed many naive neural architectures in MSE/MAE. This suggests that when model parsimony and correct exogenous specification are achievable, classical econometric models remain valuable.
- **TCN, LSTM, GRU, and hybrid models form a mid-performing cluster** with similar behavior and similar biases.
- **VAR underperforms**, confirming that linear models fail to capture commodity-driven nonlinearities.
- **Smoothing baselines (SMA, EMA)** serve as useful controls but lag trends substantially.
- The results highlight the importance of architectures capable of learning long-range nonlinear structure from commodity-driven signals.

7 Conclusion

This study examined whether commodity price information can enhance the forecasting of commodity-linked currencies, with a specific focus on the South African Rand (ZAR). By constructing a unified monthly dataset of commodity prices and exchange rates, applying econometric tests, and benchmarking a diverse set of forecasting models, we evaluated the predictive value embedded in commodity-driven signals.

Across all models, **N-BEATS emerged as the strongest performer**, achieving comparable lower error metrics and the highest directional accuracy. Unlike recurrent and convolutional sequence models, which tracked short-term fluctuations but systematically underestimated upward movements, N-BEATS successfully captured both the long-term trend and the local variations in the ZAR exchange rate. This performance difference highlights the importance of architectures capable of modelling long-range nonlinear dependencies.

The findings lead to three major conclusions. First, commodity movements contain meaningful predictive information for exchange-rate dynamics, but their contribution is highly nonlinear. Second, feature selection is important: including lagged target values together with top Granger-selected commodities (e.g., Beef, Urea, Natural Gas Europe, Platinum, TSP) significantly improved performance. Third, the choice of model architecture is critical; basis-expansion models such as N-BEATS are far better suited to capturing complex temporal relationships than traditional models or standard deep learning models (RNN, LSTM, GRU, CNN, or TCN).

Overall, the results demonstrate that commodity-informed exchange-rate forecasting is feasible, but requires sufficiently expressive nonlinear models. The superiority of N-BEATS suggests that modern deep forecasting architectures offer substantial advantages over both classical econometric approaches and conventional sequence networks.

8 Future Work

Several directions can further extend this research:

- Incorporating multi-horizon forecasts to evaluate long-term predictive stability.
- Using higher-frequency data (daily or weekly) to capture short-term market reactions.
- Exploring transformer-based architectures such as the Temporal Fusion Transformer or PatchTST.
- Applying regime-switching or volatility-aware models to capture structural breaks.
- Investigating transfer learning across commodity-linked currencies (e.g., AUD, NOK, CAD).
- Conducting walk-forward validation and stability analysis across different economic periods.

References

[1] Yıldırım, O., Erdem, E., and Kaya, S. 2022. Time-varying volatility spillovers between real exchange rates and real commodity prices for emerging market economies. *Energy Policy*. https://doi.org/10.1016/j.enpol.2022.112889.

[2] Belasen, A., and Demirer, R. 2018. Commodity-currencies or currency-commodities: Evidence from causality tests. *Energy Economics*. https://doi.org/10.1016/j.eneco.2018.10.006.

[3] Chan, K. F., Tse, Y., and Williams, M. 2011. The relationship between commodity prices and currency exchange rates: Evidence from the futures markets. NBER Working Paper (Revised Version). https://www.nber.org/system/files/chapters/c11859/revisions/c11859.rev1.pdf.

[4] Pincheira, P., and Hardy, N. 2019. Forecasting aluminum prices with commodity currencies. *Resources Policy*. https://doi.org/10.2139/ssrn.3511564.

[5] Pincheira, P., and Hardy, N. 2018. Forecasting base metal prices with commodity currencies. *Econometric Modeling: Capital Markets – Forecasting eJournal*. https://doi.org/10.2139/ssrn.3095448.

[6] Esangbedo, M., Taiwo, B., Abbas, H., Hosseini, S., Sazid, M., and Fissha, Y. 2024. Enhancing the exploitation of natural resources for green energy: LSTM-based meta-model for aluminum price forecasting. *Resources Policy*. https://doi.org/10.1016/j.resourpol.2024.105014.

[7] Magner, N., Hardy, N., Lavín, J., and Ferreira, T. 2023. Forecasting commodity market synchronization with commodity currencies: A network-based approach. *Entropy*. https://doi.org/10.3390/e25040562.

[8] Bazzi, M., et al. 2024. Looking Under the Hood of Commodity Currency Predictability. *SSRN Working Paper*. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4564504.

[9] Nguyen, Q. 2024. The Impact of Global Commodity Price Volatility on Exchange Rate Stability of Developing Countries: Evidence from ASEAN. *SSRN Working Paper*. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5338976.

[10] Zhang, Y., and He, Q. 2022. Commodity price effects on currencies: Sectoral and temporal differences. *Journal of International Money and Finance*. https://doi.org/10.1016/j.jimonfin.2022.102603.

[11] Tankov, P. 2022. Multi-factor models for commodity price dynamics: A comprehensive framework. *HAL Working Paper*. https://hal.science/hal-03758093v1.

[12] Zhang, Y., Dufour, J.M., and Galbraith, J.W. 2015. Exchange rates and commodity prices: Measuring causality at multiple horizons. *Journal of Econometrics*. https://doi.org/10.1016/j.jeconom.2015.05.003.

[13] Owusu, G., et al. 2022. Commodity prices drive real exchange rate volatility in emerging economies: Evidence from Ghana. *Energy Policy*. https://doi.org/10.1016/j.enpol.2023.113429.

[14] Bekiros, S., and Uddin, G. S. 2021. Commodity prices and exchange rates in developed and emerging economies: Non-linear Markov-switching evidence. *International Review of Financial Analysis*. https://www.researchgate.net/publication/351118543.

[15] Beckmann, J., Czudaj, R., and Koop, G. 2019. Forecasting exchange rates with commodity prices: A global country analysis. *The World Economy*. https://doi.org/10.1111/twec.12802.